

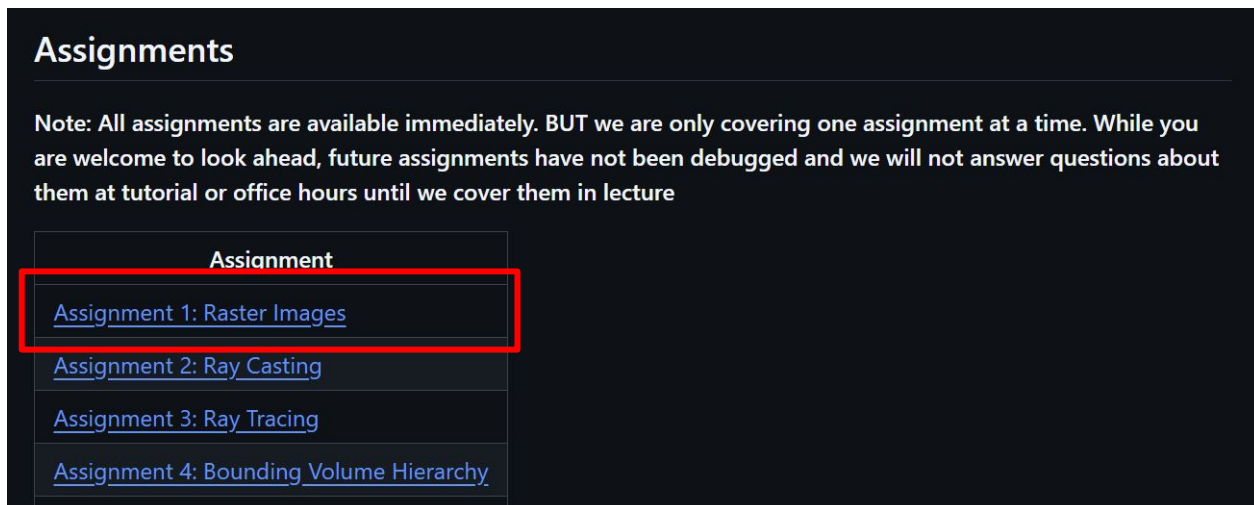
CSC317 Computer Graphics

Tutorial 1

September 4, 2025

Assignment 1: Raster Images

- Due Date: September 12, 2025 before 11:59 pm
- [Assignment](#) description can be accessed here or through course github page, under “[Assignments](#)”



Assignments

Note: All assignments are available immediately. BUT we are only covering one assignment at a time. While you are welcome to look ahead, future assignments have not been debugged and we will not answer questions about them at tutorial or office hours until we cover them in lecture

Assianment
Assignment 1: Raster Images
Assignment 2: Ray Casting
Assignment 3: Ray Tracing
Assignment 4: Bounding Volume Hierarchy

Assignment 1: Raster Images

- Every student is given ten (10) grace tokens (gt)
 - Automatically applied at midnight on the due date
 - Each grace token will provide an additional 12 hours without penalty, weekends count as late days
 - You cannot choose which assignments to apply your gt.
 - Once you run out of grace tokens, any late assignments will be counted as 0 marks.

Assignment 1: Raster Images

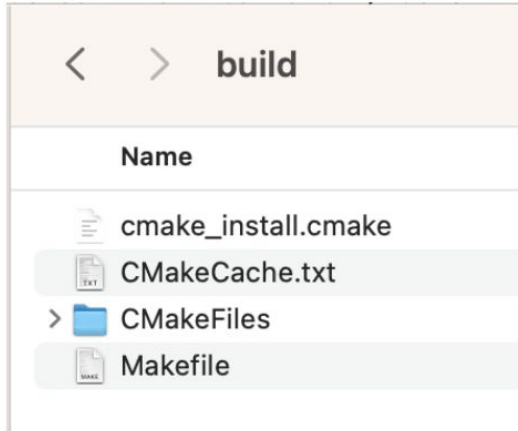
- Example: A1 is due on Friday 11:59 pm and you submit your assignment on Sunday 4 pm, you will have used 4 grace tokens.
- Friday 11:59 pm -> Saturday 11:59 am (1 gt) -> Saturday 11:59pm (1 gt) -> Sunday 11:59 am (1 gt) -> Sunday 11:59pm (1 gt).
- You will now have 6 grace tokens left for other assignments.

Assignment 1: Raster Images

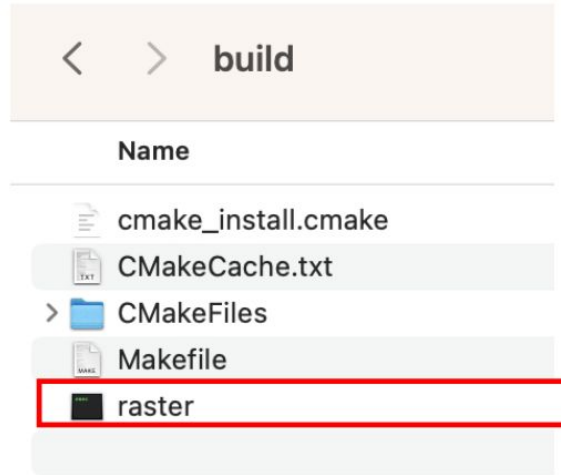
- Questions and Assistance
 - “[Issues](#)” page on github for Assignment 1
 - Email the TAs: csc317@cs.toronto.edu
- Office Hour ([Zoom](#)): 9/12 (Fri): 1:00 pm ~ 3:00 pm
 - Will not debug code for you
 - Mostly with Q&A format
- Regarding regrade requests: posted after grade release

Assignment 1: Raster Images

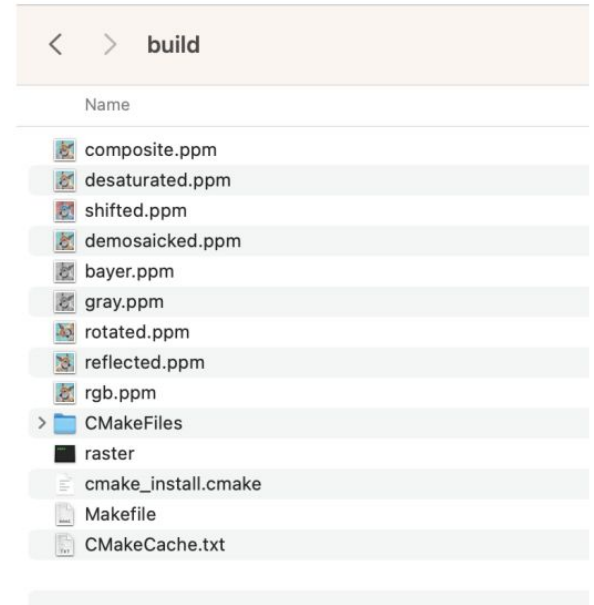
- Compilation and Running (Mac/Linux)



After “cmake ..”



After “make”



After “./raster”

Assignment 1: Raster Images

- Compilation and Running (Windows)
 - Make sure [cmake](#) and [Visual Studio](#) build tools are installed
 - Might need to restart for fresh installs

Assignment 1: Raster Images

- Compilation and Running (Windows)

```
PS C:\Users\USER\Desktop\New folder\test> git clone --recursive https://github.com/ohnooj/computer-graphics-raster-images.git
Cloning into 'computer-graphics-raster-images'...
remote: Enumerating objects: 328, done.
remote: Counting objects: 100% (105/105), done.
remote: Compressing objects: 100% (49/49), done.
remote: Total 328 (delta 67), reused 64 (delta 46), pack-reused 223 (from 1)
Receiving objects: 100% (328/328), 12.03 MiB | 16.70 MiB/s, done.
Resolving deltas: 100% (168/168), done.
PS C:\Users\USER\Desktop\New folder\test> cd .\computer-graphics-raster-images\
PS C:\Users\USER\Desktop\New folder\test\computer-graphics-raster-images> mkdir build
```

Directory: C:\Users\USER\Desktop\New folder\test\computer-graphics-raster-images

Mode	LastWriteTime	Length	Name
d-----	9/1/2025 03:47 PM		build

```
PS C:\Users\USER\Desktop\New folder\test\computer-graphics-raster-images> cd build
PS C:\Users\USER\Desktop\New folder\test\computer-graphics-raster-images\build> cmake ..
-- Building for: Visual Studio 17 2022
-- The C compiler identification is MSVC 19.44.35215.0
-- The CXX compiler identification is MSVC 19.44.35215.0
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Check for working CXX compiler: C:/Program Files (x86)/Microsoft Visual Studio/2022/BuildTools/VC/Tools/MSVC/14.44.35207/bin/Hostx64/x64/cl.exe - skipped
-- Detecting C compile features
-- Detecting C compile features - done
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Check for working CXX compiler: C:/Program Files (x86)/Microsoft Visual Studio/2022/BuildTools/VC/Tools/MSVC/14.44.35207/bin/Hostx64/x64/cl.exe - skipped
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- Configuring done (9.8s)
-- Generating done (0.0s)
-- Build files have been written to: C:/Users/USER/Desktop/New folder/test/computer-graphics-raster-images/build
```

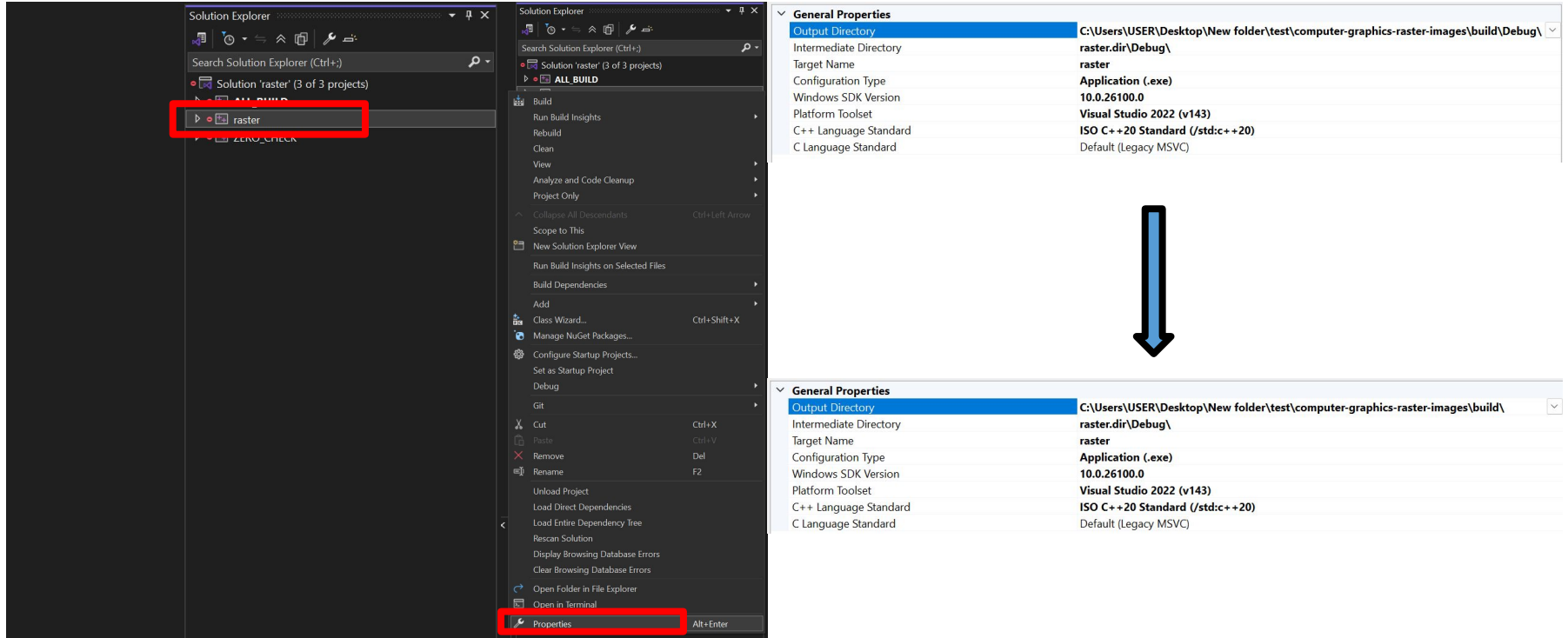
Assignment 1: Raster Images

- Compilation and Running (Windows)

 CMakeFiles	9/1/2025 03:47 PM	File folder	
 ALL_BUILD.vcxproj	9/1/2025 03:47 PM	VCXPROJ File	50 KB
 ALL_BUILD.vcxproj.filters	9/1/2025 03:47 PM	FILTERS File	1 KB
 cmake_install.cmake	9/1/2025 03:47 PM	CMake Source File	2 KB
 CMakeCache.txt	9/1/2025 03:47 PM	Text Document	15 KB
 raster.sln	9/1/2025 03:47 PM	Visual Studio Solution	4 KB
 raster.vcxproj	9/1/2025 03:47 PM	VCXPROJ File	66 KB
 raster.vcxproj.filters	9/1/2025 03:47 PM	FILTERS File	3 KB
 ZERO_CHECK.vcxproj	9/1/2025 03:47 PM	VCXPROJ File	52 KB
 ZERO_CHECK.vcxproj.filters	9/1/2025 03:47 PM	FILTERS File	1 KB

Assignment 1: Raster Images

- Compilation and Running (Windows)



The screenshot illustrates the steps to access the properties of a project in Visual Studio. In the Solution Explorer, the 'raster' project is selected. The context menu is open, and the 'Properties' option at the bottom is highlighted. A large blue arrow points from the 'Properties' option to the Properties window on the right.

Properties Window (Top):

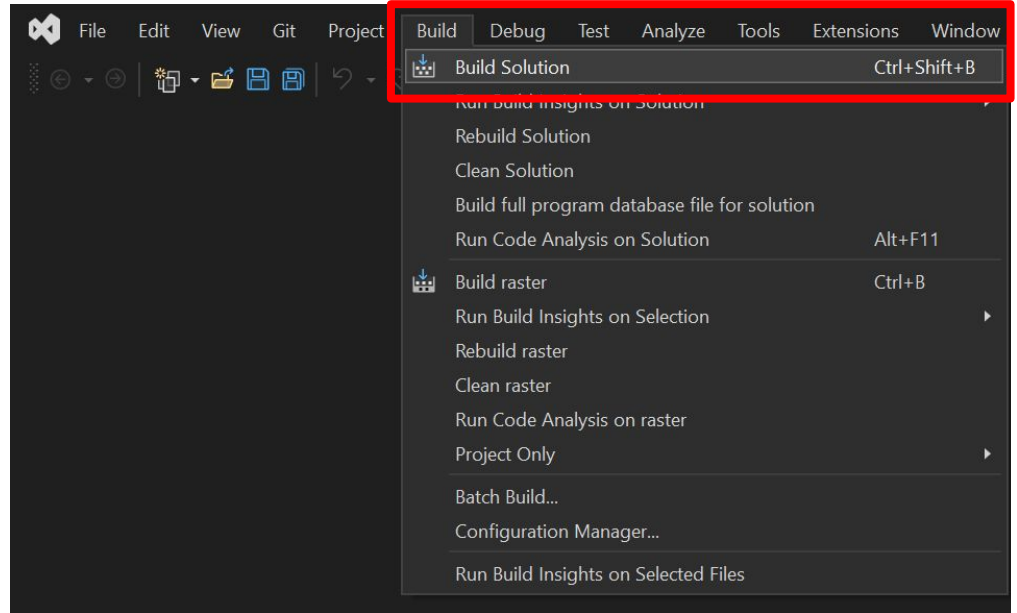
General Properties	
Output Directory	C:\Users\USER\Desktop\New folder\test\computer-graphics-raster-images\build\Debug\
Intermediate Directory	raster.dir\Debug\
Target Name	raster
Configuration Type	Application (.exe)
Windows SDK Version	10.0.26100.0
Platform Toolset	Visual Studio 2022 (v143)
C++ Language Standard	ISO C++20 Standard (/std:c++20)
C Language Standard	Default (Legacy MSVC)

Properties Window (Bottom):

General Properties	
Output Directory	C:\Users\USER\Desktop\New folder\test\computer-graphics-raster-images\build\
Intermediate Directory	raster.dir\Debug\
Target Name	raster
Configuration Type	Application (.exe)
Windows SDK Version	10.0.26100.0
Platform Toolset	Visual Studio 2022 (v143)
C++ Language Standard	ISO C++20 Standard (/std:c++20)
C Language Standard	Default (Legacy MSVC)

Assignment 1: Raster Images

- Compilation and Running (Windows)
 - Ctrl + Shift + B or



Assignment 1: Raster Images

- Compilation and Running (Windows)

```
computer-graphics-raster-images\build> ./raster
```

or

CMakeFiles	9/1/2025 03:57 PM
Debug	9/1/2025 03:58 PM
raster.dir	9/1/2025 03:57 PM
x64	9/1/2025 03:57 PM
ALL_BUILD.vcxproj	9/1/2025 03:47 PM
ALL_BUILD.vcxproj.filters	9/1/2025 03:47 PM
cmake_install.cmake	9/1/2025 03:47 PM
CMakeCache.txt	9/1/2025 03:47 PM
raster.exe	9/1/2025 03:58 PM
raster.sln	9/1/2025 03:47 PM
raster.vcxproj	9/1/2025 03:58 PM
raster.vcxproj.filters	9/1/2025 03:47 PM
raster.vcxproj.user	9/1/2025 03:55 PM
ZERO_CHECK.vcxproj	9/1/2025 03:47 PM
ZERO_CHECK.vcxproj.filters	9/1/2025 03:47 PM

Assignment 1: Raster Images

- Compilation and Running (Windows)
 - After task 2:

 composite.ppm	9/1/2025 02:25 PM	PPM File	1,669 KB
 desaturated.ppm	9/1/2025 02:25 PM	PPM File	1,695 KB
 shifted.ppm	9/1/2025 02:25 PM	PPM File	1,674 KB
 demosaicked.ppm	9/1/2025 02:25 PM	PPM File	1,424 KB
 bayer.ppm	9/1/2025 02:25 PM	PPM File	560 KB
 gray.ppm	9/1/2025 02:25 PM	PPM File	564 KB
 rotated.ppm	9/1/2025 02:25 PM	PPM File	1,674 KB
 reflected.ppm	9/1/2025 02:25 PM	PPM File	1,674 KB
 rgb.ppm	9/1/2025 02:25 PM	PPM File	1,674 KB
 raster.exe	9/1/2025 02:24 PM	Application	726 KB

Assignment 1: Raster Images

- Read [assignment handout](#) carefully:
- Program entry: main.cpp
 - Input images
 - Overall flow
- Check PPM outputs online & validate examples
- Experiment with different test cases
- Errors within a certain small tolerance range is acceptable

Assignment 1: Raster Images

- Before implementing a function, check its corresponding header file for description

```
#ifndef RGBA_TO_RGB_H
#define RGBA_TO_RGB_H

#include <vector>

// Extract the 3-channel rgb data from a 4-channel rgba image
//
// Inputs:
//   rgba width*height*4 array of 4-channel rgba instensities (i.e., rgb +
//     alpha channel for transparency)
//   width image width (i.e., number of columns)
//   height image height (i.e., number of rows)
// Outputs:
//   rgb width*height*3 array containing rgb image color intensities
//
void rgba_to_rgb(
    const std::vector<unsigned char> & rgba,
    const int & width,
    const int & height,
    std::vector<unsigned char> & rgb);

#endif
```

```
#include "rgba_to_rgb.h"

void rgba_to_rgb(
    const std::vector<unsigned char> & rgba,
    const int & width,
    const int & height,
    std::vector<unsigned char> & rgb)
{
    rgb.resize(height*width*3);
    ///////////////////////////////////////////////////
    // Add your code here
    ///////////////////////////////////////////////////
}
```

Assignment 1: Raster Images

- Task 1: `rgba_to_rgb.cpp`
 - Alpha channel: controls transparency of the pixel
 - Removing it changes the size: $w * h * 4 \Rightarrow w * h * 3$
 - Input: `const std::vector<unsigned char> & rgba`
 - Output: `std::vector<unsigned char> & rgb`
- Hint for debugging: convert to int and print to console

Assignment 1: Raster Images

- Task 2: write_to_ppm.cpp
 - Convert vector container to .ppm file
 - Needs to support both rgb and grayscale images
- PPM
 - Include header number (P2/P3)
 - Open in text editor (and compare to validation file)

Assignment 1: Raster Images

- Task 3: reflect.cpp



Assignment 1: Raster Images

- Task 4: rotate.cpp



- What is the relationship between old/new width/height?

Assignment 1: Raster Images

- Task 5: `rgb_to_gray.cpp`



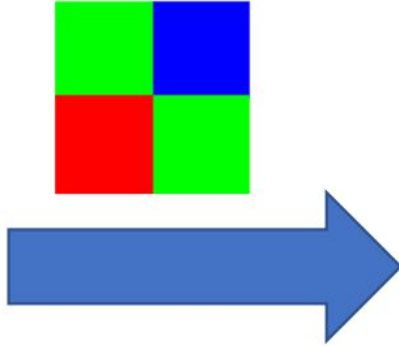
$$i = 0.2126r + 0.7152g + 0.0722b$$



- 3 channels (width * height * 3) to 1 channel (width * height)

Assignment 1: Raster Images

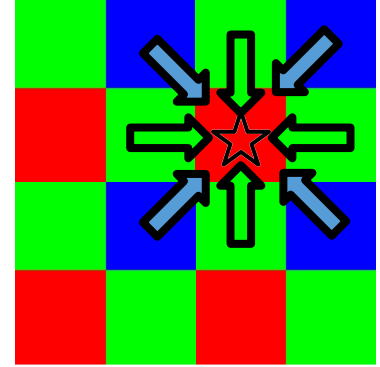
- Task 6: `simulate_bayer_mosaic.cpp`



- 3 channels (width * height * 3) to 1 channel (width * height)

Assignment 1: Raster Images

- Task 7: demosaic.cpp
 - Inverse the previous task
- For each Pixel:
 - Take the exact value of itself (use the R value if on a red)
 - Average the values for the other two channels
- Example:
 - Use the red
 - Take average of blue and green around it
- 1 channel (width * height) to 3 channels (width * height * 3)



Assignment 1: Raster Images

- Task 8: `rgb_to_hsv.cpp` & `hsv_to_rgb.cpp`
 - Helper functions for next two tasks
 - RGB to HSV
 - Check with standard values online for correctness
 - HSV to RGB
-
- Be careful with data types (i.e. inputs are of type double)
 - Be careful with input/output ranges (e.g. 0 to 1 vs. 0 to 255)

Assignment 1: Raster Images

- Task 9: hue_shift.cpp
 - Use the two helper functions



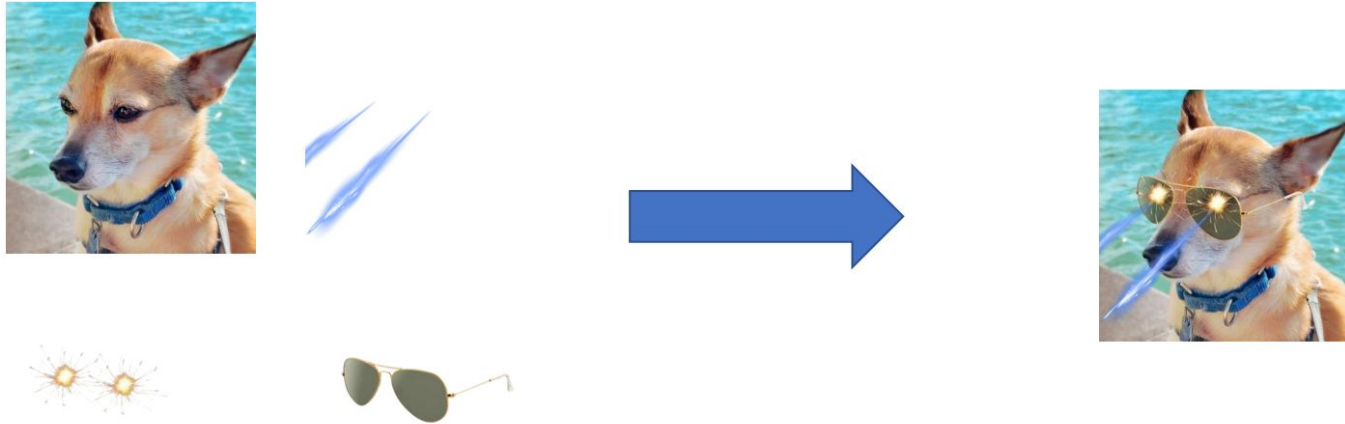
Assignment 1: Raster Images

- Task 10: desaturate.cpp
 - Use the two helper functions



Assignment 1: Raster Images

- Task 11: over.cpp
 - Implement the Over Layering operation



- Hint: Read [Porter/Duff Compositing and Blend Modes](#)

Assignment 1: Raster Images

- Submitting Assignments:
 - Where? Markus
 - What? All .cpp files in the src directory (also listed on Markus)

demosaic.cpp

desaturate.cpp

hsv_to_rgb.cpp

hue_shift.cpp

over.cpp

reflect.cpp

rgb_to_gray.cpp

rgb_to_hsv.cpp

rgba_to_rgb.cpp

rotate.cpp

simulate_bayer_mosaic.cpp

write_ppm.cpp
